# Gallery Description Cards

### Week 7 Assignment - Problem 2 (Basic)

ArtPiece is abstract and assigns every object a unique final ID in its constructor. Painting and Sculpture directly extend it and provide different descriptions.

## Java solution

```
abstract class ArtPiece {
    private static int nextId = 1001;
    private final String pieceId;
    ArtPiece() { pieceId = "ART-" + nextId++; }
    public abstract String describe();
    String getPieceId() { return pieceId; }
}
class Painting extends ArtPiece {
    private final String title;
    Painting(String title) { this.title = title; }
    public String describe() {
        return "Painting: " + title + ", framed on canvas";
    }
}
class Sculpture extends ArtPiece {
    private final String title;
    Sculpture(String title) { this.title = title; }
    public String describe() {
        return "Sculpture: " + title + ", carved from stone";
    }
}
public class Main {
    public static void main(String[] args) {
        Painting p = new Painting("Sunset Fields");
        Sculpture s = new Sculpture("The Thinker II");
        System.out.println(p.describe());
        System.out.println(s.describe());
        System.out.println(p.getPieceId());
        System.out.println(s.getPieceId());
    }
}
```

## Sample output

```
Painting: Sunset Fields, framed on canvas
Sculpture: The Thinker II, carved from stone
ART-1001
ART-1002
```

Note: The prompt does not specify an ID prefix, so this solution uses ART-1001, ART-1002, and so on.